

Laboratoriya 15

Mavzu: JSON formatda ma'lumotlarni olish va ulash.

Kirish:

Hozirgi kunda internet orqali ma'lumot almashishning asosiy qismi API (Application Programming Interface) texnologiyalari orqali amalga oshiriladi. Turli xil dasturlash tillari va platformalar (masalan, mobil ilova va server) o'zaro "gaplashishi" uchun umumiy til kerak. JSON (JavaScript Object Notation) aynan shu vazifani bajaradi. U matn ko'rinishidagi format bo'lib, ham kompyuterlar o'qishi uchun qulay, ham insonlar tushunishi uchun oson.

Ish maqsadi:

JSON strukturasi bilan yaqindan tanishish va uning sintaksisini o'rganish.

Dasturlash tili (masalan, Python yoki JavaScript) yordamida JSON fayllarni o'qish (parsing) va yozish (serialization) ko'nikmalarini shakllantirish.

Tashqi API'lardan (masalan, ob-havo yoki valyuta kurslari) JSON formatida ma'lumot olishni amalda sinab ko'rish.

Nazariy qism:

JSON nima?

JSON — ma'lumotlarni saqlash va tashish uchun mo'ljallangan yengil format. Garchi nomi JavaScript bilan bog'liq bo'lsa-da, u tildan mustaqil va deyarli barcha zamonaviy dasturlash tillari tomonidan qo'llab-quvvatlanadi.

JSON ma'lumotlar turlari:

JSON quyidagi ma'lumot turlarini o'z ichiga oladi:

Object (Obyekt): Jingalak qavslar ichida { } kalit va qiymat juftligi.

Array (Massiv): Kvadrat qavslar ichida [] tartiblangan ro'yxat.

String: Qo'shtirnoq ichidagi matn.

Number: Sonlar (butun yoki o'nlik).

Boolean: true yoki false.

Null: Bo'sh qiymat.

JSON bilan ishlashning asosiy bosqichlari:

Serialization (Ulash): Dasturdagi obyekt yoki lug'atni (dictionary) JSON formatidagi matnga o'tkazish.

Deserialization (Olish/Parsing): Kelgan JSON matnini dastur tushunadigan obyektga (masalan, Python lug'atiga) aylantirish.

Misol:

Oddiy JSON ma'lumoti quyidagicha ko'rinadi:

JSON

{

 "ism": "Asadbek",

```
"yosh": 21,  
"talabami": true,  
"fanlar": ["Matematika", "Dasturlash"]  
}
```

Amaliy qism:

Index.html kod:

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <title>Modern Filterable Table</title>  
  <!-- Google Fonts -->  
  <link rel="preconnect" href="https://fonts.googleapis.com">  
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>  
  <link  
href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700&display=swap"  
rel="stylesheet">  
  <!-- jQuery -->  
  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.7.1/jquery.min.js"></script>  
  <!-- Custom CSS -->  
  <link rel="stylesheet" href="style.css">  
</head>  
<body>  
  <div class="container">  
    <header>  
      <h2>Filterable Table</h2>  
      <p class="subtitle">Search through posts by title, body, or ID</p>  
    </header>  
    <div class="search-container">  
      <svg class="search-icon" xmlns="http://www.w3.org/2000/svg" width="20" height="20"  
viewBox="0 0 24 24"  
      fill="none" stroke="currentColor" stroke-width="2" stroke-linecap="round"  
stroke-linejoin="round">  
        <circle cx="11" cy="11" r="8"></circle>  
        <line x1="21" y1="21" x2="16.65" y2="16.65"></line>  
      </svg>  
      <input id="myInput" class="search-input" type="text" placeholder="Search  
posts...">  
    </div>  
    <div class="table-wrapper">  
      <table>  
        <thead>  
          <tr>  
            <th>User ID</th>  
            <th>ID</th>  
            <th>Title</th>  
            <th>Body</th>
```

```

        </tr>
    </thead>
    <tbody id="myTable">
        <!-- Data will be populated here -->
    </tbody>
</table>
<div id="loader" class="loader-container" style="display: none;">
    <div class="loader"></div>
</div>
<div id="error"></div>
</div>
</div>
<!-- Custom Script -->
<script src="script.js"></script>
</body>
</html>

```

Style.css:

```

:root {
    --primary-color: #4f46e5;
    /* Electric Indigo */
    --primary-hover: #4338ca;
    /* Deep Indigo */
    --accent-color: #8b5cf6;
    /* Vivid Violet */
    --bg-color: #f0f9ff;
    /* Light Sky Blue Background */
    --card-bg: #ffffff;
    --text-primary: #0f172a;
    /* Slate 900 */
    --text-secondary: #475569;
    /* Slate 600 */
    --border-color: #e2e8f0;
    --row-hover: #e0e7ff;
    /* Light Indigo Hover */
    --header-bg: linear-gradient(135deg, #4f46e5 0%, #8b5cf6 100%);
    --shadow-sm: 0 1px 2px 0 rgb(0 0 0 / 0.05);
    --shadow-md: 0 10px 15px -3px rgb(0 0 0 / 0.1), 0 4px 6px -2px rgb(0 0 0 / 0.05);
    --radius: 1rem;
}
* {
    box-sizing: border-box;
    margin: 0;
    padding: 0;
}
body {
    font-family: 'Inter', sans-serif;
    background-color: var(--bg-color);
    color: var(--text-primary);
    line-height: 1.5;
    padding: 2rem;
}

```

```
}  
  
.container {  
  max-width: 1200px;  
  margin: 0 auto;  
}  
  
header {  
  margin-bottom: 2rem;  
  text-align: center;  
}  
  
h2 {  
  font-size: 2.5rem;  
  font-weight: 800;  
  background: var(--header-bg);  
  -webkit-background-clip: text;  
  -webkit-text-fill-color: transparent;  
  margin-bottom: 0.5rem;  
  letter-spacing: -0.025em;  
}  
  
p.subtitle {  
  color: var(--text-secondary);  
  font-size: 1rem;  
}  
  
.search-container {  
  margin-bottom: 1.5rem;  
  position: relative;  
  max-width: 400px;  
  margin-left: auto;  
  margin-right: auto;  
}  
  
.search-input {  
  width: 100%;  
  padding: 0.75rem 1rem 0.75rem 2.5rem;  
  border: 1px solid var(--border-color);  
  border-radius: var(--radius);  
  font-size: 1rem;  
  transition: all 0.2s;  
  outline: none;  
  box-shadow: var(--shadow-sm);  
}  
  
.search-input:focus {  
  border-color: var(--primary-color);  
  box-shadow: 0 0 0 3px rgba(99, 102, 241, 0.2);  
}  
  
.search-icon {
```

```

    position: absolute;
    left: 0.75rem;
    top: 50%;
    transform: translateY(-50%);
    color: var(--text-secondary);
    pointer-events: none;
}

.table-wrapper {
    background-color: var(--card-bg);
    border-radius: var(--radius);
    box-shadow: var(--shadow-md);
    overflow: hidden;
    border: 1px solid var(--border-color);
}

table {
    width: 100%;
    border-collapse: collapse;
    text-align: left;
}

thead {
    background: var(--header-bg);
}

th {
    padding: 1.25rem 1rem;
    font-weight: 600;
    color: #ffffff;
    text-transform: uppercase;
    font-size: 0.875rem;
    letter-spacing: 0.05em;
}

td {
    padding: 1rem;
    border-bottom: 1px solid var(--border-color);
    color: var(--text-primary);
    font-size: 0.875rem;
}

tr:last-child td {
    border-bottom: none;
}

tbody tr {
    transition: background-color 0.2s;
}

tbody tr:hover {

```

```

        background-color: var(--row-hover);
    }

    /* Loading Spinner */
    .loader-container {
        display: flex;
        justify-content: center;
        padding: 2rem;
    }

    .loader {
        border: 4px solid var(--border-color);
        border-top: 4px solid var(--primary-color);
        border-radius: 50%;
        width: 40px;
        height: 40px;
        animation: spin 1s linear infinite;
    }

    @keyframes spin {
        0% {
            transform: rotate(0deg);
        }

        100% {
            transform: rotate(360deg);
        }
    }

    #error {
        color: #ef4444;
        text-align: center;
        margin-top: 1rem;
    }

    /* Responsive */
    @media (max-width: 768px) {
        body {
            padding: 1rem;
        }

        .table-wrapper {
            overflow-x: auto;
        }
    }
}

```

Script.js kod:

```

$(document).ready(function () {
    const $tableBody = $("#myTable");
    const $loader = $("#loader");
    const $error = $("#error");

```

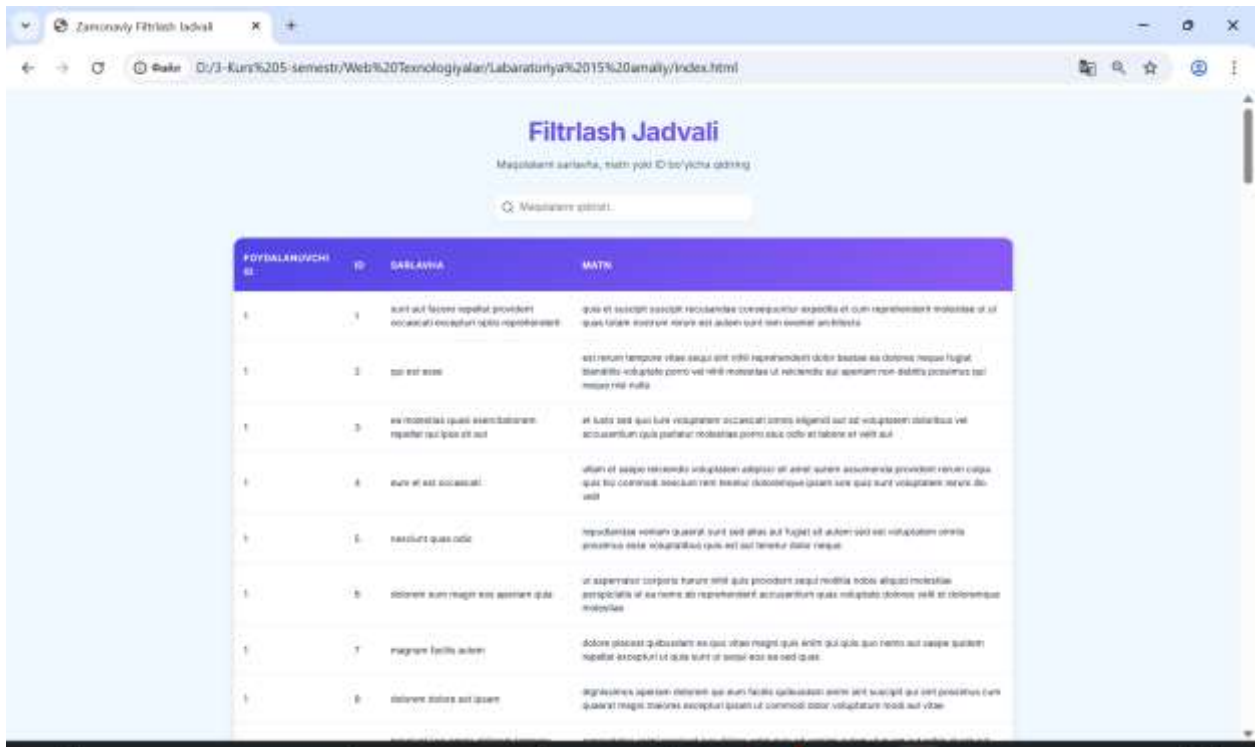
```

const $searchInput = $("#myInput");
let allData = [];
// Fetch data
$.ajax({
  url: "https://jsonplaceholder.typicode.com/posts",
  type: "get",
  beforeSend: function() {
    $loader.show();
    $tableBody.hide();
  },
  success: function (data) {
    allData = data;
    renderTable(data);
    $loader.hide();
    $tableBody.fadeIn();
  },
  error: function (err) {
    $loader.hide();
    $error.html("Error loading data. Please try again later.");
    console.error("Error:", err);
  }
});
// Search functionality
$searchInput.on("keyup", function() {
  const value = $(this).val().toLowerCase();
  const filteredData = allData.filter(item => {
    // Search in title, body, and ID
    return (
      item.title.toLowerCase().includes(value) ||
      item.body.toLowerCase().includes(value) ||
      item.id.toString().includes(value) ||
      item.userId.toString().includes(value)
    );
  });
  renderTable(filteredData);
});
function renderTable(data) {
  $tableBody.empty();
  if (data.length === 0) {
    $tableBody.append(`
      <tr>
        <td colspan="4" style="text-align: center; padding: 2rem; color: var(--text-secondary);">
          No results found
        </td>
      </tr>
    `);
  }
  return;
}
let rows = "";
$.each(data, function (k, v) {

```

```
rows += `<tr>
    <td>${v.userId}</td>
    <td>${v.id}</td>
    <td>${v.title}</td>
    <td>${v.body}</td>
</tr>`;
});
$tableBody.append(rows);
}
});
```

Natija:



Fihriyash Jaadvai			
Maqolalarni sarlavhasi, ma'ni yoki ID bo'yicha qidirish			
🔍 ID			
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